BC180408070: AJLAL ABBAS

MTH622 Quiz 3

Quiz Start Time: 09:50 AM, 0

Question # 10 of 1 (at time: 09:57:40 AM, 05 August 2021)

If angular momentum $\Omega = 2\omega \sin \omega t$ then the torque τ is

Select the correct option

Reload M

- $2\omega^2 \cos \omega t$ Ans
- $-2 \cos \omega t$

Click to S... Answer & Move t...

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Derivation

Kinetic Energy

The kinetic energy for a particle is given by the following scalar equation:

$$T = \frac{1}{2}mv^2$$

Where:

- T is the kinetic energy of the particle with respect to ground (an inertial reference frame)
- m is the mass of the particle
- v is the velocity of the particle, with respect to ground

Kinetic Energy of a Rigid Body

For a rigid body experiencing planar (two-dimensional) motion, the kinetic energy is given by the following general scalar equation:

$$T = \frac{1}{2}mv_c^2 + \frac{1}{2}I_c\omega^2 \qquad (1)$$

where the first term in equation (1) shows the kinetic energy due to the motion of center of mass and second term shows the rotational kinetic energy and subscript c denotes the center of mass and v_c denotes the velocity of c.m and I_c denotes the inertia matrix w.r.t c.m.

Kinetic Energy of a Rigid Body w.r.t Origin

If the rigid body is rotating about a fixed point O that is attached to ground, we can express the kinetic energy as:

$$T = \frac{1}{2}I_o\omega^2$$

Where:

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Question # 9 of 10 (Start time: 09:57:09 AM, 05 August 2021)

If P is any point on the ellipsoid of inertia, then the moment of inertia of a rigid body about any line $\vec{OP}$ is equals to the reciprocal of the square of the length of $|\vec{OP}|$

Select the correct option

- ○ true
- ○ false

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Question # 8 of 10 (Start time: 09:56:25 AM, 05 August 2021)

If an object is thrown into the air, different parts of the object can follow quite complicated paths, but the center-of-mass will follow ____

Select the correct option

- a parabola
- a straight line

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Quiz Start Time: 09:50 AM

Question # 2 of 10 (Start time: 09:51:43 AM, 05 August 2021)

If the center of mass is at origin then the coordinate axes (xyz axes) and the principal axes of the rigid body are

Select the correct option

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- perpendicular
- coincident

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Quiz Start Time: 09:50 AM, 05 August 2021

Question # 7 of 10 (Start time: 09:55:24 AM, 05 August 2021)

Total Marks:

Consider a right circular cone of density ρ. radius a and height h is composed of elementary circular discs of small thickness each parallel to the base of the cone. Choose the z-axis as the axis of symmetry and consider a typical disc of radius r and width δz them mass of the disc is

Select the correct option

Reload Math Equations

- $\delta m = \rho \pi r^2 \delta z$
- $\delta m = \rho 2\pi r \delta z$

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$T = \frac{1}{2} I_0 \omega^2$



$T = \frac{1}{2} m v_c^2 + \frac{1}{2} I_c \omega^2$

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Question # 5 of 10 (Start time: 09:54:13 AM. 05 August 2021)

If the axis of rotation passes through
the center of mass of the rigid body then it is called

Select the correct option

- spin
- revolution

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Question # 4 of 10 (Start time: 09:52:58 AM, 05 August 2021)

The moment of inertia of a particle about y-axis is

Select the correct option

○	$I_{yy} = m(x^2 + y^2)$
○	$I_{yy} = mxz$

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MTH622 Quiz 3

Quiz Start Time: 09

Question # 3 of 10 (Start time: 09:52:23 AM, 05 August 2021)

Kinetic energy T of a rigid body in a planar motion, is defined by

Select the correct option

- $T = \frac{1}{2} I_0 \omega^2$
- $T = \frac{1}{2} m v_c^2 + \frac{1}{2} I_c \omega^2$

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Time Left 84 SECS

MTH622:Quiz 3

Quiz Start Time: 09:50 AM, 05 August 2021

Question # 1 of 10 (Start time: 09:50:09 AM, 05 August 2021)

Total Marks: 1

If a particle of mass 4 moving with velocity $4\hat{i}$ collides another particle of mass 5 with velocity $3\hat{i}$. Then what will be the velocity of first particle after collision if the velocity of second particle after collision is $2\hat{i}$.

Select the correct option

Reload Math Equations

- $\frac{21}{4}\hat{i}$
- $\frac{15}{4}\hat{i}$

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